Decentralization and Criminal Gangs in El Salvador: Impacts on Municipal Finances and Local Economic Development

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Abstract

Recent literature has argued that decentralization can create new forms of leverage for criminal gangs, affecting their territorial presence and levels of violence. This paper reverses the causal arrow and analyzes how gangs affect the performance of decentralized institutions. In particular, we study the case of El Salvador, a country with ubiquitous presence of gangs, and find that higher presence of gangs reduced municipalities' fiscal revenues, therefore increasing their dependence on transfers from the central government. This result is mostly found in mid-size municipalities, as the effect is not significant for small and large municipalities. In addition to depressing revenues, gang presence was also associated with lower municipal spending, particulary in the case of capital spending and in small muncipalities, and less service provision. We also find that gangs undermine economic activity, not just in the municipalities where they operate but also through spillover effects in neighboring jurisdictions.

JEL Codes: H0; O0; R0
Keywords: gangs, decentralization, municipal finance, El Salvador

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Abbreviations

ARENA	Alianza Republicana Nacionalista party
COMURES	Corporation of Municipalities of the Republic of El Salvador
EHPM	Multi-purpose Household Survey or Encuesta de Hogares de Propósitos Múltiples
FLMN	Frente Farabundo Martí para la Liberación Nacional party
FODES	El Salvador's Fund for Municipal Economic and Social Development or Fondo para el Desarrollo Económico y Social; a source of transfers from the central government to muncipal governments
GS2LS estimator	generalized spatial two-stage least squares estimator
ISDEM	Salvadorian Institute of Municipal Development or Instituto Salvadoreño de Desarrollo Municipal
MS-13	Mara Salvatrucha, a large criminal organization; one gang (or maras) in El Salvador, along with Barrio 18
SIGET	El Salvador's General Superintendence of Electricity and Telecommunications or Superintendencia General de Electricidad y Telecomunicaciones

1. Introduction

Can decentralization promote local development in countries with a significant presence of non-state armed actors? Scholars have engaged in a robust debate on this topic. According to the more-optimistic strand of the literature, decentralization can help reduce violence, as it helps address some of the grievances that led non-state actors to initiate their violence in the first place. For instance, decentralization can facilitate the tailoring of policies to fit the particular ethnic and/or cultural practices of subnational regions (Bland, 2006; Brinkheroff, 2011). Decentralization can also give rebel organizations the opportunity to transform themselves into political parties, win local elections, and take control of subnational governments that now have real resources and responsibilities (Bermeo, 2002; Stepan, 2001). Decentralization can even reduce the stakes associated with control of the central state, reducing incentives to rebel.

Another strand of literature is less optimistic, arguing that decentralization can lead to increased violence, hurting local development. There are several reasons this might occur. For instance, in decentralized environments, the different levels of government often fail to act cohesively as a single decision-making body, enabling criminal organizations to grow and increase levels of violence (Durán-Martínez, 2018; Rios, 2013). Another reason decentralization can hurt regional development is that it can exacerbate conflicts if it empowers regionally oriented political parties that foster secessionist sentiment (Brancati, 2006). Decentralization can also allow rebel groups to penetrate subnational governments and use decentralized resources to engage in service delivery, thereby enhancing their legitimacy as they continue to try to overthrow the central state (Eaton, 2006).

In this paper, we propose a novel argument for how decentralization and gang violence might be interrelated, hurting local development. We argue that gangs can undermine the ability of decentralized institutions to function as intended, hurting state capacities, and therefore economic development. In particular, the territorial presence of gangs can limit local fiscal autonomy because the rents they extort from local populations have a depressing effect on the ability of municipalities to collect their own revenues, as citizens who are

extorted by gangs had fewer resources to comply with their local fiscal obligations. In addition to depressing tax revenues, the presence of gangs can reduce municipal spending and limit service provision.

We test the argument presented above in El Salvador. El Salvador is an interesting case study as decentralization played a key role in helping the country overcome the legacies of its deadly civil war, a twelve-year conflict (1980–1992) during which armed rebel organization unsuccessfully sought to overthrow a national government dominated by a narrow set of economic elites. While these elites continued to control the national government in the years that followed the conflict through the electoral success of the Alianza Republicana Nacionalista (ARENA) party, the rebels transformed their organization into a political party that was able to win elections in a growing number of municipalities, including eventually in the capital city of San Salvador, and the national government (Holland, 2016). Electoral rules regarding municipalities that guaranteed a majority for the mayor's party meant that those municipalities under the control of the Frente Farabundo Martí para la Liberación Nacional (FLMN) party became meaningful spaces of political opposition vis-à-vis ARENA at the national level.

Meanwhile, many local governments experimented with the use of innovative participatory mechanisms to broaden decision making beyond partisan politicians (Bland, 2011), and fiscal decentralization in the form of more-generous revenue sharing gave all municipal governments a greater share of fiscal revenues. The transformation of local governments into critical spaces of political contestation laid the foundations for the FMLN's eventual presidential victories in 2009 and 2014 along with the practice of ideologies alternating in power. El Salvador's relatively stable two-party system has since been transformed by the rise of Nayib Bukele's newly hegemonic Nuevas Ideas (New Ideas) party, but decentralization nevertheless deserves much of the credit for the political advances that El Salvador made in its emergence from internal armed conflict, namely the transformation of a rebel group into a competitive political party.

While decentralization helped a non-state armed actor transform itself into a political party that eventually won the presidency, El Salvador's experience with decentralized governance was subsequently undermined by a non-state armed actor of a very different sort: criminal gangs. Gangs (or maras)

date back to the 1960s in El Salvador, but the problem significantly worsened when the United States returned approximately 40,000 convicted criminals to El Salvador after the end of the civil war, including Salvadorans who had joined either the Mara Salvatrucha (MS-13) or Barrio 18 in Los Angeles (Sviatschi, 2022).[1] Over time, and partially in response to the adoption of hard-line (mano dura) policies by both ARENA and FMLN governments, gangs have transformed from "loose neighborhood-level federations that operated largely autonomously and lacked internal governing bureaucracies" into "hierarchically organized structures with organizational bases inside the country's prisons" (Moncada, 2021, 91). With 60,000 members operating in El Salvador, gangs have operated in almost all of the 262 municipalities, where weaknesses in the rule of law enabled them to engage in a wide range of extortion activities. In particular, criminal taxation became a "regular, costly, and humiliating facet of everyday life" (Moncada, 2021, 96).

This report shows that gangs negatively impacted the ability of decentralization to deliver on its promise in El Salvador, with several effects standing out. In one in particular, related to revenue collections, we find that territorial presence of gangs limited the capacity of municipalities to collect their own revenues. This might be related to citizens having lower incomes and abilities to pay as a consequence of the rents gangs extracted from them. This negative effect of gangs on municipal own-revenue collections is seen in mid-size municipalities, as no statistically significant effect is found for small and large municipalities.

In addition to these negative fiscal outcomes, we find that the presence of gangs was associated with lower municipal spending and limited service provision. The drop in municipal spending was particularly strong in capital spending and in small municipalities. Along with these negative fiscal outcomes, we find that gangs undermined economic activity, not just in the municipalities where they operate but through spillover effects in neighboring jurisdictions as well. This might be related to gangs inhibiting factor mobility between municipal jurisdictions, through activities such as limiting labor flows

[1] MS-13 and Barrio 18 are the two main gangs in El Salvador, and their origins go back to East Los Angeles, in California. Sviatschi (2022) refers to these two main gangs, but there are other gangs that are splinters from those two gangs.

due to neighborhood checkpoints and extorted payments on the transport of goods.

This paper is organized into five sections, including this introduction. Section 2 briefly reviews the theoretical literature on decentralization in the context of non-state armed actors. Section 3 describes the design of decentralization in El Salvador in the period before President Bukele came to power in 2019, along with the territorial expansion of gangs across Salvadoran municipalities. Section 4 explores the fiscal performance of municipalities across several dimensions, including local taxation, spending, and debt. Section 5 provides conclusions by assessing the impact of gangs on fiscal and economic outcomes, demonstrating that gangs have an especially corrosive effect on the fiscal performance of mid-size municipalities, along with negative effects on service provision and local economic activity. The appendixes include support and elaborations in the form of maps, tabular information, definitions of variables, and sources.

2. Decentralization and Non-State Armed Actors

In many countries of the Global South, the state does not enjoy a monopoly on the legitimate use of force, and instead its attempt to control the national territory is contested by a range of challengers who themselves are able to exert territorial control (Luna and Giraudy, 2017). Scholars have engaged in a robust debate on the merits of decentralization in these contexts. According to the more-optimistic arguments in the literature, decentralization can address some of the grievances that have led non-state actors to initiate rebellion in the first place, including a lack of responsiveness on the part of national governments in faraway capitals, and the inability of subnational governments to tailor policies to fit the particular ethnic and/or cultural practices of subnational regions (Bland, 2006; Brinkheroff, 2011). Decentralization can generate a number of other positive effects in these environments by lowering the stakes associated with control of the central state and by giving rebel organizations the opportunity to transform themselves into political parties able to win power over subnational governments that now have real resources and responsibilities (Bermeo, 2002; Stepan, 2001).

Other scholars have been more cautionary and argue that decentralization can backfire. According to Dawn Brancati (2006), decentralization is likely to exacerbate conflicts if it empowers regionally oriented political parties that stand to gain electorally by further fanning the flames of grievances, including secessionist sentiment. In other cases, rebel groups can penetrate subnational governments and use decentralized resources to engage in service delivery, thereby enhancing their legitimacy as they continue to try to overthrow the central state (Eaton, 2006).

Not all non-state armed actors seek to overthrow the state or secede from it, however. Criminal gangs, in sharp contrast to rebels, insurgents, and guerrillas, pose a different set of challenges. The concern is not that gangs use decentralized resources to strengthen the drive to secede from the country or to bolster their prospects of defeating the government militarily. In contrast, in a less-visible but potentially more-pervasive dynamic, gangs can disrupt the various causal mechanisms that have led theorists to posit a positive relationship between decentralization and improved developmental outcomes.

One of the main arguments for decentralization is that it enables subnational governments to compete with one another through the mechanism of factor mobility (Oates, 1972, 2005; Tiebout, 1961). Labor and capital may more easily flow between neighboring subnational jurisdictions than they can between countries, and voters might be able to "vote with their feet" by opting to relocate to local jurisdictions where policy offerings better match their preferences (Ter-Minassian, 1997). Competition between subnational governments for capital investment and human resources can encourage local politicians to adopt the kinds of "market-preserving" policies that enhance the country's overall prospects for economic growth (Weingast, 1995).

However, through a number of illicit behaviors, gangs can limit factor mobility by extracting payoffs from economic actors seeking to transport goods between municipalities, and by imposing checkpoints that limit the mobility of local residents. Criminal organizations interact with businesses in ways that powerfully affect the local economic environment in which they operate (Fenizia, 2018), not just in illegal markets but in formal markets as well, where they can affect levels of production, employment, and prices (Ferrante,

Fontana, Reito, 2021; Montoya, 2016; Rozo, 2014). In some cases, scholars have documented salary variations that are due chiefly to restrictions imposed on the mobility of less-skilled labor in localities with a high gang presence; higher labor costs due to these restrictions may result in higher production costs (Roso, 2014).

Gangs can also disrupt another important mechanism through which decentralization is intended to improve fiscal outcomes, which is the ability of local governments to collect own-source revenues. When fiscal decentralization takes the form of devolving tax bases to subnational governments (rather than sharing revenues that are generated through centrally collected taxes), it can have a host of positive impacts for accountability, encouraging local taxpayers to demand that their local governments use own-source revenues to provide services (Azfar et al., 1999). Voters may have an easier time monitoring how local governments use the taxes that the voters pay to those local governments than is the case for taxes owed to the central government. At the same time, the political and administrative costs associated with collecting unpopular local taxes can lead to halfhearted or under-resourced local tax efforts, discouraging local authorities from collecting the tax bases they have been assigned. While insufficient local tax effort deserves the attention it has received from scholars of decentralization, it is important to note that widespread gang activity may also be impeding efforts by local authorities to collect taxes. Impoverished residents may be unable to pay "rent" to the criminal gangs who are directly and violently extorting them, and to pay the municipal taxes and fees they are legally obligated to pay. If both gangs and mayors (along with the rest of each local government) are trying to tap the same resources, there are good reasons to expect that gangs will win out.

When gangs can inhibit the collection of local taxes and fees by local governments, they can trigger a number of other negative fiscal outcomes identified by the literature on decentralization. For example, the inability of local governments to collect local taxes is likely to worsen the government's dependence on fiscal transfers from the center (Rodden, Eskeland, Litvack, 2003). A large body of literature has emphasized the perverse incentives that transfer systems can generate, especially when they take the form of nonearmarked revenues that local governments consider to be "blank checks"

(Bahl and Martinez-Vazquez, 2006; Inman and Rubinfeld, 1997). Transfer systems can also increase the scope for politically oriented discretion on the part of the national government. In addition to worsening the reliance on transfers, the inability to collect local revenues because criminal gangs are tapping the same resource base may lead to increased levels of local debt and pressure for bailouts (Di Cataldo and Mastrorocco, 2021; Rodden et al., 2003).

If criminal gangs can obstruct local tax collection and increase the reliance on transfers, they can also powerfully affect the relationship between decentralization and service provision by local governments. Here scholars have pointed to two problematic dynamics. The first is that the presence of gangs may make it more difficult for neighboring municipalities to use rival approaches to service provision as a means of differentiating themselves from one another (as anticipated by theories of fiscal federalism).

The second is that gangs can also inhibit the mobility of factors of production between municipal jurisdictions through their capacity to control territory and establish checkpoints that act as toll taxes on the movement of people, public transportation, and goods and services (Brown, Montero, Schmidt-Padilla, and Sviatschi, 2020; Melnikov, Schmidt-Padilla, and Sviatschi, 2021). Di Cataldo and Mastrorocco (2020) argue that municipalities where organized crime has taken hold experience a distortion in the provision of public services due to a variety of rent-seeking activities. Bueno (2022) and Córdova (2017) likewise attribute significant impacts on service provision to gang activity. In addition to shaping the mix of service provision, gangs can impact the overall level of service as well. Several scholars have also documented how organized crime reduces the incentives for local governments to provide critical goods and services across the board and their capacity to do so, starting with the protection of citizens (Acemoglu, De Feo, De Luca, 2020; Arends, 2021; Beraldo et al., 2017; Melnikov et al., 2021; Pinotti, 2015). Indeed, regarding one especially salient mechanism of local government, a government's perceived need to provide services to ingratiate itself with local residents may be depressed when the municipality is being held captive by gangs (Beraldo et al., 2017).

3. The Evolution of Decentralization, Local Finances, and Gang Violence

This section presents the main features of fiscal decentralization arrangements and some municipal fiscal results in El Salvador. Appendix 1 and 2 summarize the extent of gang violence across Salvadorean municipalities.

3.1 The Institutional Structure of Municipal Governance in El Salvador

Reflecting its unitary status and highly centralized past, El Salvador maintained until 2023 just one level of subnational government composed of 262 municipalities, and these vary significantly in population, economic dynamism, and levels of urbanization. At the intermediate level, El Salvador is organized into 14 departments whose function it is to represent the central state and whose authorities (governors) are appointed by the national government rather than elected in their own right. Although decentralization occurred just at the municipal level of government, there were three dimensions of change: political, administrative, and fiscal.[2]

In the political sphere, the electoral rules for municipal elections allow parties to present closed lists that are topped by the mayoral candidate and followed by as many candidates as there are seats on the municipal council (Córdova and España-Nájera, 2021). Prior to the 2015 municipal electoral reform, the winning party was guaranteed complete control of the municipal council, with no opposition seats. Since then, opposition parties have been allowed to gain representation on the council, but the mayor's party is still guaranteed a majority. Despite the highly partisan and majoritarian design associated with municipal governance, political decentralization also led to the emergence of bipartisan bargaining and coordination among mayors within the Corporation of Municipalities of the Republic of El Salvador (COMURES), which sought to defend vis-à-vis the center the roles played by municipalities (Bland, 2011).

[2] The municipal code assigns a number of other key roles to municipalities, including trash collection; municipal policing; the construction of housing; the regulation of local transport, commercial activities, and restaurants; and the promotion of industrial development, education, and health (Vargas, 2022).

In addition to providing for elections, the 1980 municipal code in El Salvador provides for a number of other mechanisms to enhance participation, including local development councils and open assemblies (cabildos abiertos) several times a year. As Bland argues, "local government allowed for alternation in executive power and demonstrated that turnover of government from one party to another was possible without inciting violence" (Bland, 2011, 866).

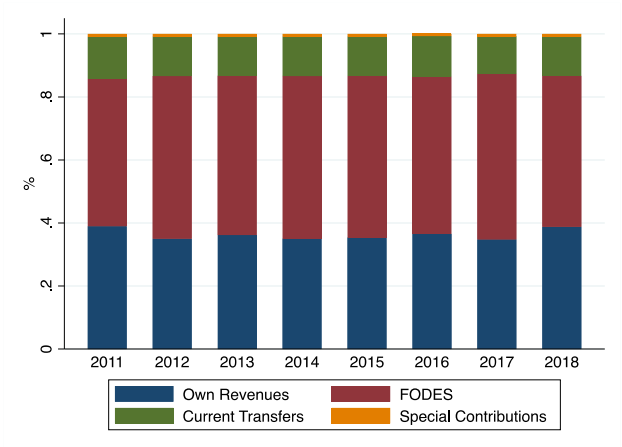
El Salvador is unusual in Latin America in that it denies municipalities the ability to levy property taxes.[3] Instead municipal governments can collect direct fee-based taxes paid by individuals and companies that develop economic activities within the district. Municipalities can also raise non-tax revenues from user fees and charges, as well as revenue from the sale of goods and services. In addition to those own-source revenues, municipalities received transfers through a revenue-sharing system that was developed beginning in 1988 in the form of the Fund for Municipal Economic and Social Development (Fondo para el Desarrollo Económico y Social or FODES). FODES funds are distributed according to population, equity, poverty, and land area criteria, and the municipalities that receive the funds are required to devote 80 percent of the revenues to capital investment projects and no more than 20 percent to operating expenses.

3.2 Fiscal Decentralization Characteristics

During the period 2010–2018, two aspects characterized the municipal finances in El Salvador: (i) high reliance on intergovernmental transfers, and (ii) low generation of own-source revenues. At an aggregate level, FODES transfers represented 48 percent of total municipal revenues during the analyzed period. Municipalities' own-source revenues represented a smaller share, accounting on average for about 37 percent of total municipal revenues (see Figure 1). Most of these own-source revenues correspond to user fees levied on commercial, industrial, financial, and service activities.

[3] Historically, El Salvador has lacked a property tax for municipal governments. Several efforts to create such a tax have been made and none succeded: (i) in the late 1960s, supported by experts from Spain's Ministry of the Interior; and (ii) in the period between the late 1990s and in the early 21st century, supported by the United States Agency for International Development (USAID).

Figure 1. Sources of Municipal Revenues, 2011–2018



Source: Authors' elaboration based on Portal de Transparencia Fiscal—El Salvador; ISDEM.

However, the aggregate figures hide large heterogeneities across the municipalities of different sizes (see Table 1), which are categorized by population as small (less than 10,000), medium (10,000 to 50,000) , and large (greater than 50,000). In particular, small municipalities are highly dependent on FODES transfers; in 2018, they received 10 times more FODES transfers than their own revenues (see Table 1). Medium-size municipalities collect relatively more own-source revenues than smaller municipalities, and so their ratio of FODES to own-source revenues is about a third of that observed for smaller municipalities. Finally, for large municipalities, on average FODES transfers are only 20 percent higher than their own revenues.

Table 1. Per Capita Tax Collection and Size of FODES Transfers to Municipalities, 2011–2018

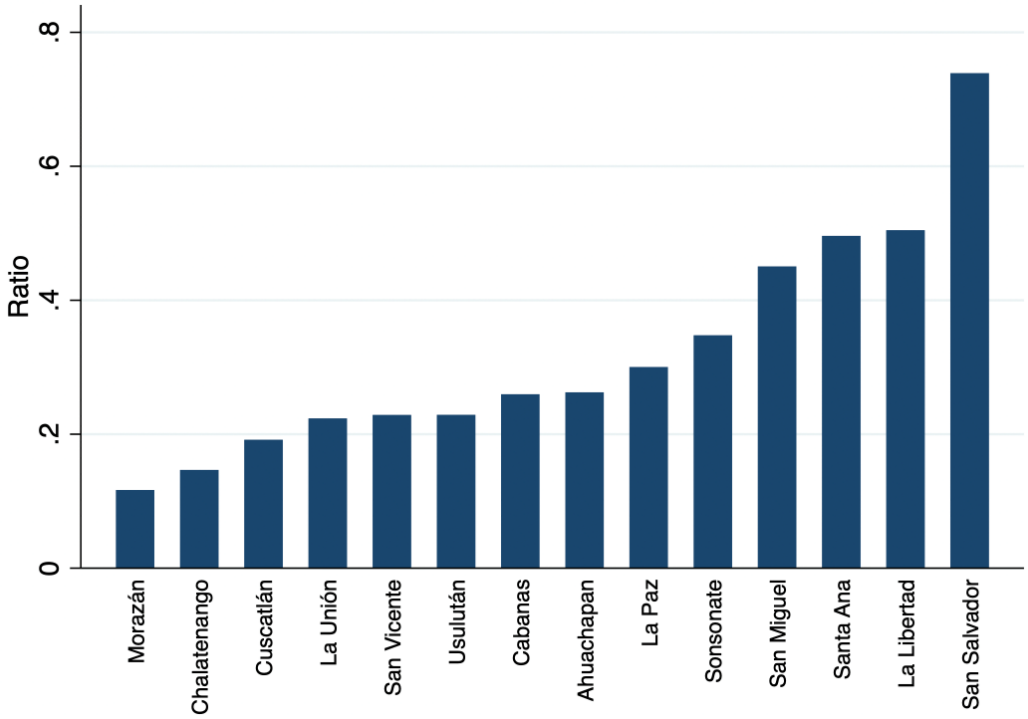
	Small Municipalities Population <10K		Medium Municipalities Population 10K—50K		Large Municipalities Population >50K	
Year	Tax collection (US$ per capita)	Ratio of FODES to own revenues	Tax collection (US$ per capita)	Ratio of FODES to own revenues	Tax collection (US$ per capita)	Ratio of FODES to own revenues
2011	13.8	25.0	27.7	4.2	35.1	1.2
2012	13.9	13.7	29.0	5.1	36.5	1.4
2013	15.7	12.4	31.9	4.7	39.0	1.3
2014	16.6	13.5	31.1	4.5	41.4	1.2
2015	17.1	14.5	33.0	4.5	43.9	1.2
2016	18.4	13.0	35.0	4.2	46.3	1.2
2017	19.2	11.1	34.2	3.8	46.2	1.1
2018	21.5	10.0	39.0	3.5	51.5	1.2

Source: Authors' elaboration.

In aggregate, for 2018, the average expenditure of municipalities was US$159 per capita.[4] On average, 32 percent of local expenditures were financed by own-source revenues. However, given the differences in municipalities by size and economic activity, the greater the population size, the greater the share of own-source revenues that finance municipal expenditures. Figure 2 shows low budgetary autonomy for municipalities in most departments, but that municipalities in the most-populated and developed departments of El Salvador (namely, San Miguel, Santa Ana, La Libertad, and San Salvador) are less dependent on FODES transfers than other municipalities.

[4] In terms of local expenditure, the main functions of municipalities are: removing solid waste; ensuring safety in local markets, parks, and local streets; overseeing construction and maintenance of cemeteries; fostering local economic development; and supporting community housing services.

Figure 2. Budgetary Autonomy: Municipalities Aggregated by Department, 2018



Source: Authors' elaboration based on Portal de Transparencia Fiscal—El Salvador; ISDEM. *Note:* In this figure, ratio refers to the proportion of the municipal budget from own-source revenue expressed as a decimal figure. Thus, only for the municipalities in San Salvador do own-source revenues exceed 0.6 or 60 percent.

4. Fiscal and Economic Impacts of Gang Violence at the Local Level

To measure the impact of gangs on municipal governments' municipal finances, capacity to deliver services, and local economic activity, we developed a panel data model with fixed effects. The model is estimated using data from 258 municipalities in El Salvador for the years 2011–2018. We analyze the effects on fiscal outcomes as well as on local economic activity.

4.1 Methodology for Effects on Fiscal Outcomes

We evaluated the effect of gang activities on different fiscal outcomes variables.

The econometric specification used was a panel data model of the form:

$$Y_{i,t} = \beta_0 + \beta_1 Gangs_{i,t-1} + \beta_k X_{i,t} + \sum_{t=1}^{T-1} \gamma_t D_t + a_i + \mu_{i,t}$$

Where $Y_{i,t}$ is the dependent variable that will be, according to each model, the variables: (i) own-source revenue per capita, (ii) expenditure per capita, and (iii) provision of public services. The variable represents the presence of the gangs as measured by the number of homicides per thousand inhabitants attributed to MS-13 per municipality.

The number of homicides per thousand inhabitants allows us to measure the presence of gangs as well as the intensity of criminal activities. To mitigate any potential endogeneity problem, particularly concerning the simultaneity and reverse causality between the presence of gangs and fiscal outcomes, we employ a one-period lagged variable of gang presence. This approach assumes that current fiscal outcomes are influenced by past gang activities rather than being contemporaneously determined, thereby reducing the likelihood of simultaneity bias.[5]

The term β_0 denotes the intercept term, which captures the average value of the dependent variable across all municipalities and across time. Term β_1 is the coefficient of the variable of interest. The specification also includes a vector

[5] Endogeneity concerns were addressed through the application of different specifications using instrumental variables (IV) and other diagnostic tests to ensure they do not pose an issue in our estimation. In this process, we found no significant problems beyond potential bidirectional causality, which has been accounted for with the inclusion of the lagged variable.

with other control variables $X_{i,t}$ such as percentage of urban population, population density, average household income, average energy consumption, and investment in capacity building within local governments, among others.

Additionally, we include a vector with time dummy variables (D_t), representing different time periods. Finally, we include a vector a_i capturing individual-specific random effects. Random effects are included under the assumption that individual entities have unique characteristics that remain constant over time and may influence the relationship and dynamics between the dependent variable ($Y_{i,t}$) and other non-observed independent variables such as, for example, the institutional strength of each municipality, which are heterogeneous. In this context, while examining fiscal outcomes and the presence of gangs, random effects could encapsulate unobservable factors such as local governance dynamics, economic conditions, social and cultural norms, historical events, institutional variations, and demographic characteristics unique to each jurisdiction.

We also explore heterogeneous effects by analyzing the impact of gangs in municipalities of different sizes. This is relevant as the existing literature identifies a direct and positive relationship between city size and the incidence of criminal activities (Gaigné and Zenou, 2015; Glaeser and Sacerdote, 1999; Kahn, 2010). The hypothesis behind this is that given that larger municipalities (area size and population) have a greater level of financial autonomy and diverse economic structure, their finances are less vulnerable to the gangs' territorial control and illegal activities, and therefore they have the capacity to secure funding for the provision of public services. Thus, gangs have a lesser impact on the capacity of large municipalities to collect own revenues. To allow for heterogeneous effects, we estimated the model splitting the sample by three groups: (i) municipalities with a population of less than 10,000 inhabitants, which is a value close to the series' median (see Table 2), (ii) municipalities with a population between 10,000 and 50,000 inhabitants, and (iii) municipalities with a population greater than 50,000 inhabitants. The population distribution of El Salvador is asymetric. The median population size of a Salvadoran municipality is 10,076, and those municipalities with 10,000 inhabitants or less represent 50 percent of all municipalities. Municipalities in that bracket are classified as small. Ten percent of the municipalities are large and have more than 50,000 inhabitants, and the remaining 40 percent of

municipalities have a population distribution between 10,000 and 50,000 inhabitants, see (Table 2).

Table 2. Main Statistics of Variable Population by Municipality, 2019

Statistics		Percentiles	
Mean	22,238	10%	3,158
Std. deviation	38,047	25%	5,486
Variance	1.45E+09	50%	10,076
Skewness	5	75%	21,675
Kurtosis	28	90%	52,359
Observations	245	99%	241,403

Source: Authors' elaboration.

4.2 Methodology for Effects on Local Economic Activity

To identify the effects of gangs on local economic activity as well as measure whether gang activities in a municipality have spillover effects in adjacent municipalities, we developed a model with economic activity as a dependent variable. We used the level of energy consumption (in kilowatt-hours, kWh) in the locality as a proxy of economic activity. To measure the impact of gang activities at the municipal level and the uncertainty gangs generate, we included a cross-sectional spatial autoregressive model, considering an adjustment estimator using a generalized spatial two-stage least squares (GS2LS estimator) following the procedure developed by Kelejian and Prucha (1998).[6] The specification is as follows:

$$Y_{i,t} = \beta MaraSalva_{i,t} + \rho W_{y;i,t} + \delta W_{X;i,t} + \gamma Z_{i,t} + \varepsilon_{i,t}$$

$$\varepsilon_{i,t} = \lambda W_{\varepsilon} + \mu_{i,t}$$

Here $Y_{i,t}$ is energy consumption used as a proxy variable for economic activity measured as millions of kWh. The variable represents the presence of gangs using both the homicides from MS-13 and homicides from all gangs (measured as homicides per thousand inhabitants) in different models. To avoid the

[6] Panel data could not be used, given that the spatial autocorrelation models in any given period only function with a balanced panel, and the availability of data is limited for the variables considered.

possibility of endogeneity, we considered a lagged value of the variable. Additionally, $Z_{i,t}$ is a vector that includes other control variables, among which are the level of intensity of street lighting at night,[7] the urban population, family incomes, per capita expenditure, and the percentage of the population over age 65.

The matrix *W* represents a spatial weights matrix, which contains the spatial arrangements of the different localities to be analyzed and each entry represents the spatial weight associated with locality *i* and locality *j*; in this case if two municipalities share a border, the value is equal to 1, and otherwise 0. Due to this specification, the matrix allows the spatial measurement of different variables. The specification includes three types of interactions that incorporate a spatial component captured by the *W* matrix, that is: (i) an endogenous interaction, which represents the effect of the dependent variable (economic activity) on the economic activity of other municipalities, captured by the estimator ρ, which measures the effect on the economic activity (kWh consumption) in neighboring municipalities; (ii) the exogenous effect captured by the estimator δ of the variable, which measures the effect of gangs' presence in the economic activity of neighboring localities and which is the variable that allows testing whether there is an impact of the gangs' illegal actions on the economic activity of neighboring localities; and (iii) the last interaction, included in the error vector, which captures other unobserved variables with some degree of precision. In the specification, W_{ε} represents other unobserved variables with some degree of spatial structure in neighboring localities.

4.3 Data

A database was constructed for 2011 to 2018,[8] using different data sources (see Appendix 3). Most fiscal variables come from the Fiscal Transparency Portal (El Portal de Transparencia Fiscal de la El Salvador), administered by the El

[7] Data for nighttime lighting was obtained from Google Earth Engine and the Administrative Map from the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) Latin America and the Caribbean. This variable is constructed using satellite recognition of faint levels of brightness on the Earth's surface during the night, which reveals a city's level of nighttime luminosity. This variable differs from the measurement of energy consumption, which is an indicator of economic activity (which takes place mostly during the daytime). It is worth remembering, however, that both variables may be correlated to the existence of urban centers.

[8] Subsequent years were not considered due to the possible distortion that may have been caused to different variables by the effects of the COVID-19 pandemic.

Salvador Finance Ministry (Ministerio de Hacienda). Data on FODES come from the Salvadorian Institute of Municipal Development (Instituto Salvadoreño de Desarrollo Municipal, ISDEM). Meanwhile, data on homicides come from the National Police Open Data (Data Abiertos de la Policía Nacional). The control variables come from different sources. Electricity consumption at the municipal level comes from annual bulletins of the General Superintendence of Electricity and Telecommunications (Superintendencia General de Electricidad y Telecomunicaciones or SIGET), whose annual bulletins provided the information regarding electricity consumption at the municipal level, used as a proxy variable for economic activity. Meanwhile, local night luminosity comes from Google Earth. Information regarding the socioeconomic situation of Salvadorian households comes from the Multi-purpose Household Survey (Encuesta de Hogares de Propósitos Múltiples, EHPM), which is representative of the entire country. Appendix 3 presents more information regarding the variables employed.

4.4 Impacts

Effects on fiscal management

We find that gangs had a significant impact on local finances in El Salvador. Gangs had an effect on the level of revenue collection, a result seen most in medium-sized municipalities. A negative effect is also found on municipal spending. The results below consider the effects of MS-13. Similar effects are obtained when running the regressions considering the presence of other gangs.

Effects on own-source revenues

In the specification using own revenues as the dependent variable and violence by MS-13 as the main independent variable, we found a higher intensity of gang presence significantly reduces own-revenue collections. We find that an increase by 1 homicide per 1,000 inhabitants by MS-13 leads to a reduction in per capita revenue by US$1.1, with the results being statistically significant at a 10 percent confidence level (see Table 3).

Table 3. Effects of MS-13 Violence on Per Capita Own Revenue

	(1)	(2)	(3)	(4)
Variables	All municipalities	Population < 10K	Population 10K–50K	Population > 50K
Gangs	-1.136*	-0.426	-5.613*	-22.287
	(0.829)	(0.623)	(2.864)	(17.779)
Kw_mm	0.217	0.133*	0.853*	0.148**
	(0.125)	(0.179)	(0.435)	(0.054)
UrbanPopul	-0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Transf_pc	-0.130*	-0.015	-0.095	0.740
	(0.049)	(0.012)	(0.092)	(0.627)
Density	-0.016	-0.056	0.245	6.105
	(0.129)	(0.161)	(0.982)	(7.532)
Garbage_sv	2.204	0.451	10.871***	17.144
	(1.971)	(1.341)	(3.853)	(19.793)
Constant	27.078***	15.305***	9.079	-5.850
	(5.487)	(3.172)	(7.049)	(26.326)
Observations	1,308	465	646	197

Source: Authors' elaboration.
Notes: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Column 2 of Table 3 shows that small municipalities do not have a statistically significant impact on own-source revenue per capita due to gang activity. These small municipalities had low own-source revenue per capita and were highly dependent on FODES transfers. Thus, their financial flow seems to be less influenced by violence within their jurisdiction. About 130 municipalities have less than 10,000 inhabitants, and within this group, the average per capita collection is US$32, indicating that the majority of them are below average. Additionally, within the same group, the average per capita transfers amount to US$170 for the same year (2018).

In the case of the medium-sized municipalities (see Table 3, column 3), we found that the presence of MS-13 has a negative and significant impact on the municipalities' own-source revenues per capita. Medium-size municipalities' own-source revenues experienced the greatest drop due to illegal gang activities, to US$5.8 per homicide per 1,000 inhabitants.

Finally, in large municipalities, there are no statistically significant impacts of gang activity on per capita own-source revenues. Possible explanations are that these municipalities (including big cities such as San Salvador, Santa Tecla, Santa Ana, and San Miguel) are important economic centers that host a diverse group of industries, commercial activities, and financial services. Revenues in these municipalities tend to be more robust compared with in the rest of the municipalities, allowing the large municipalities to better withstand the negative impact of gang violence. In contrast, medium-sized municipalities are more fragile and have less-diversified economies, mostly comprised of small- and medium-sized businesses that are more vulnerable to the local social and economic context. Additionally, medium-sized municipalities may have limited resources to contain and reduce gang activities.

Furthermore, a complementary analysis was conducted, considering the effects of violence from other gangs besides MS-13, with the aim of identifying distinct impacts on the level of local per capita revenue. According to the data presented in Appendix 4, the presence of other gangs has no effects on local per capita revenue.

Effects on local expenditures

We found that the presence of MS-13 has a general significant and negative impact on municipal spending per capita. Specifically, 1 homicide per 1,000 inhabitants by MS-13 leads to a reduction of US$17.9 in per capita spending (Table 4). This effect is most pronounced in the smallest municipalities, where 1 homicide per 1,000 inhabitants due to gang activities leads to a reduction of US$20 in per capita spending. When we include “other gangs” as an interest variable, results are quite similar, showing the negative effect of gang presence on local expenditure, particularly in small municipalities (see Appendix 5).

Table 4. Effect of the Mara Salvatrucha on Expenditures

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	-17.694*** (6.347)	-20.112** (12.785)	-0.489 (7.180)	-38.447 (29.544)
Kw	-0.006 (0.084)	-1.788* (1.590)	0.137 (0.00)	-0.033 (0.020)
UrbanPopul	-0.000*** (0.000)	0.001 (0.001)	-0.000 (0.000)	-0.000** (0.000)
Total_revenue_pc	1.077*** (0.210)	2.454*** (0.611)	0.891*** (0.218)	0.675*** (0.184)
Density	8.661* (2.687)	7.798* (2.425)	-4.102 (2.848)	-24.177 (16.557)
Constant	85.488*** (9.396)	50.137 (29.655)	84.611*** (5.641)	35.412*** (4.288)
Observations	1331	479	655	197

Source: Authors' elaboration
Notes: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * p < 0.05, ** p < 0.01, *** p < 0.001.

For robustness, additional estimations were included to determine the MS-13 effect on current expenditure (which includes expenditure on goods and services, and expenditure on personnel and social obligations) and capital expenditure (local public investment) in per capita terms. These specifications are presented in Table 5. As seen in overall results, the impact on current and capital expenditure is significant and negative. When we disaggregated current and capital expenditures, we found that with 1 additional homicide per 1,000 inhabitants, the drop in capital expenditures is much larger than on current spending. While capital spending drops by US$19.7, current expenditures drops by US$2.9.

The strong negative effect on capital expenditure might be related to fear of gang reprisals, which can also hinder the smooth execution of public works, affecting project timelines and escalating costs. These results may suggest a lack of territorial control, which is consistent with the literature

characterizing the social and institutional effects of gangs, where gang domination of territory limits the entry of people or personnel not recognized or approved by the gangs who control the territory. This situation affects municipal and nonmunicipal employees whose work is critical for creation, maintenance, and service provision. Córdova (2017) associates gang-controlled territory with higher levels of poverty and a lower level of municipal-type services. The impact on current expenditures can be associated with greater difficulties in maintaining operating capacity for public service provision (due to the local control of the territory by gangs), and therefore, less coverage by public services and/or lower quality of public services.

These findings highlight the need to consider the size of the locality when studying the impact of gang-related activities on different types of expenditures. The results suggest that small-sized localities are particularly affected by the presence of the gangs, leading to significant reductions of expenditure. Further research is necessary to explore the underlying factors contributing to these varying effects across different population sizes.

Table 5. Effect of the Mara Salvatrucha on Expenditures by Municipality Size

	Effect on current expenditures				Effect on capital expenditures			
	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	-2.866**	-3.884**	2.426	-15.151	-19.702***	-16.949*	-5.381	-9.386
	(1.058)	(1.193)	(2.328)	(13.248)	(4.639)	(7.578)	(7.044)	(26.668)
Kw	0.076***	-0.587	0.133***	0.008	-0.060	-1.388	-0.111	-0.028
	(0.017)	(0.324)	(0.036)	(0.013)	(0.058)	(0.940)	(0.088)	(0.024)
Urban_popul	-0.000***	-0.000	0.000	-0.000	-0.000***	0.001	-0.001***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Transf_pc	0.557***	0.367***	0.526***	0.764***	0.352***	1.884***	0.133	0.429***
	(0.025)	(0.041)	(0.040)	(0.036)	(0.072)	(0.189)	(0.068)	(0.071)
Density	0.625**	0.580*	-0.315	-6.328	6.150***	7.117***	-2.580	-13.676
	(0.230)	(0.237)	(0.829)	(5.975)	(1.032)	(1.470)	(2.560)	(12.242)
Constant	23.018***	35.489***	19.515***	13.220***	70.934***	25.025*	73.683***	24.461***
	(1.604)	(2.921)	(2.808)	(3.021)	(4.949)	(11.479)	(4.093)	(5.795)
sigma_u								
Observations	1331	479	655	197	1331	479	655	197

Source: Authors' elaboration.
Note: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Effects on public service provision and municipal debt

We tested the relationship between the presence of MS-13 and the provision of public services, measured as the percentage of inhabitants who have access to a municipal public waste collection service. We found a significant and negative correlation between the presence of MS-13 and public service provision, particularly in medium-size municipalities.[9] Results show that, for each additional homicide per 1,000 inhabitants, the coverage of public services drops by 0.2 percent. The model does not show any impact on public service provision for small and large municipalities. Also, the model does not yield results that indicate that gangs had an impact on a municipality of any size by increasing its debt (Table 6).

Table 6. Effect of the Mara Salvatrucha on Municipal Debt by Municipality Size

Variable	All municipalities	Population < 10K	Population 10K–50K	Population > 50K
Maras	-0.002* (0.001)	0.004 (0.006)	-0.002* (0.001)	0.000 (0.000)
Urban_popul	0.000*** (0.000)	0.000 (0.000)	0.000** (0.000)	0.000* (0.000)
Household_income	0.000*** (0.000)	0.000** (0.000)	0.000*** (0.000)	-0.000 (0.000)
PC_own_rev	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.001)
Density	0.001 (0.006)	0.000 (0.006)	0.002 (0.010)	-0.012 (0.028)
MunCapacBuild	0.025 (0.015)	0.049* (0.023)	-0.010 (0.020)	-0.032 (0.020)
Constant	0.194*** (0.042)	0.205** (0.070)	0.085* (0.037)	0.517*** (0.059)
Observations	1391	520	662	209

Source: Authors' elaboration.
Note: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

[9] The effect of gangs on public service in other size municipalities is marginally significant with a level of confidence less than 10 percent.

Complementary estimation was conducted considering the number of homicides associated with other gangs. The results are similar in terms of the impact in medium-size municipalities (Appendix 6).

Effects on local economic dynamics

This section explores the impact of gangs' extortion and forced payments by firms and people on economic activity at the municipal level. We find a significant negative effect on local economic activity and spillover effects to neighboring localities. Results in Table 7 indicate that for every homicide attributed to MS-13 and to gangs in general that occurs in one locality (results for the last specification is shown in Appendix 7), the consumption of electricity within the local territory is reduced by 3.1 million kWh and it reduces the consumption in neighboring localities by 5.2 million kWh. Considering that the average consumption per municipality in a year is 29 million kWh, this impact represents a reduction of almost 10 percent on local activity.[10]

[10] It is important to consider that there are significant variations in electricity consumption among localities. In aggregate terms, the average electricity consumption per department varies, with some departments reaching as high as 120 million kWh, such as the case of San Salvador (or 972 kWh per capita), while other departments like Morazán record a consumption of 2.5 million kWh (or 256 kWh per capita).

Table 7. Impact of Mara Salvatrucha on Economic Activity

Variable	kWh
MaraSalva	-3.107** (0.996)
luminosity	0.013*** (0.001)
PobUrb	0.000 (0.000)
ingfa	0.059** (0.020)
gasto_pc	-0.005 (0.011)
agedpopulation	143861 (87197.9)
Constant	-51.824*** (13.071)
Wma= Spatial Correlation Outputs	
MaraSalva	-5.250* (2.149)
kw_mm	0.074 (0.081)
Observations	144

Source: Authors' elaboration.
Note: Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In relation to other control variables, such as nighttime luminosity and household income, there is a positive impact on local economic activity. In the first case, greater nighttime luminosity is associated with improved infrastructure and public services, such as street lighting and residents' access to transportation and electricity, which can facilitate the mobility of factors of production. Additionally, in the case of household income, localities with a higher income level are associated with greater demand for goods and services, as well as a willingness to invest capital in the development and expansion of economic activities.

5. Conclusions

This paper provides empirical evidence that fiscal results of municipal governments are negatively impacted when gangs control territory. In El Salvador, the territorial control of gangs is associated with lower municipal tax revenues, lower municipal expenditures, and decreased municipal capacity to provide public services. We found a high sensitivity in revenue collection. According to the results, for each increase by 1 in homicides by gangs per 1,000 inhabitants leads to a reduction in revenue per capita by US$1.1. This statistically significant result is mostly found in medium-sized municipalities, which, for this study, were those with 10,000 to 50,000 inhabitants. We found that in municipalities of this size, the impact of gang activities is US$5.6 for each homicide attributed to gangs per 1,000 inhabitants. In municipalities of other sizes, the effect is not statistically significant.

The presence of MS-13 also has a significant and negative impact on municipal spending per capita. Specifically, on average, for each 1 percent increase in homicides by gangs, the per capita expenditure decreases by US$17.9. Here, the greatest impact is on the smallest municipalities, where the drop in expenditures per capita is US$20 per 1 homicide per 1,000 inhabitants. When disaggregating current and capital expenditures, we found that for every extra homicide per 1,000 inhabitants, capital expenditures drop by US$19.7, while current expenditures drop US$2.9. In terms of service provision, there is a significant and negative correlation between the presence of gangs and public service provision, particularly in medium-size municipalities, where garbage collection decreased by 0.2 percent per 1,000 inhabitants due to gang presence. We also found a significant negative effect on local economic activity and spillover effects to neighboring localities. Results indicate that for every homicide attributed to gangs in general that occurs in one locality (see Appendix 8), the consumption of electricity within the local territory is reduced by 3.1 million kWh and consumption in neighboring localities is reduced by 5.2 million kWh. Considering that the average consumption per municipality in a year is 29 million kWh, this reduction represents a reduction of almost 10 percent on local economic activity.

Our research revealed that, in addition to the adverse impacts on own-source revenues, gangs also exerted inhibitory effects on the level of local

expenditure within affected municipalities. This phenomenon can be attributed to the gangs' ability to control territorial areas, allowing them to impose toll taxes on the movement of factors, including people, public transportation, and goods and services between municipal jurisdictions, a result that was also reported in Brown, Montero, Schmidt-Padilla, and Sviatschi (2020). Consequently, the local governments' capacity to spend on essential services and infrastructure is constricted, as their ability to collect taxes and generate income is hampered by gang-imposed toll taxes. The disparities observed among municipalities of different sizes necessitate further investigation into the intricate interplay between economic factors, administrative capacity, and territorial characteristics that influence municipal sensitivity to gang-related issues.

Finally, we find that gang activities had a negative impact not just on the municipality in which they operated but also in adjacent and neighboring municipalities. Understanding these dynamics is crucial for devising effective strategies to address the financial challenges posed by gangs and restore the economic growth of affected municipalities.

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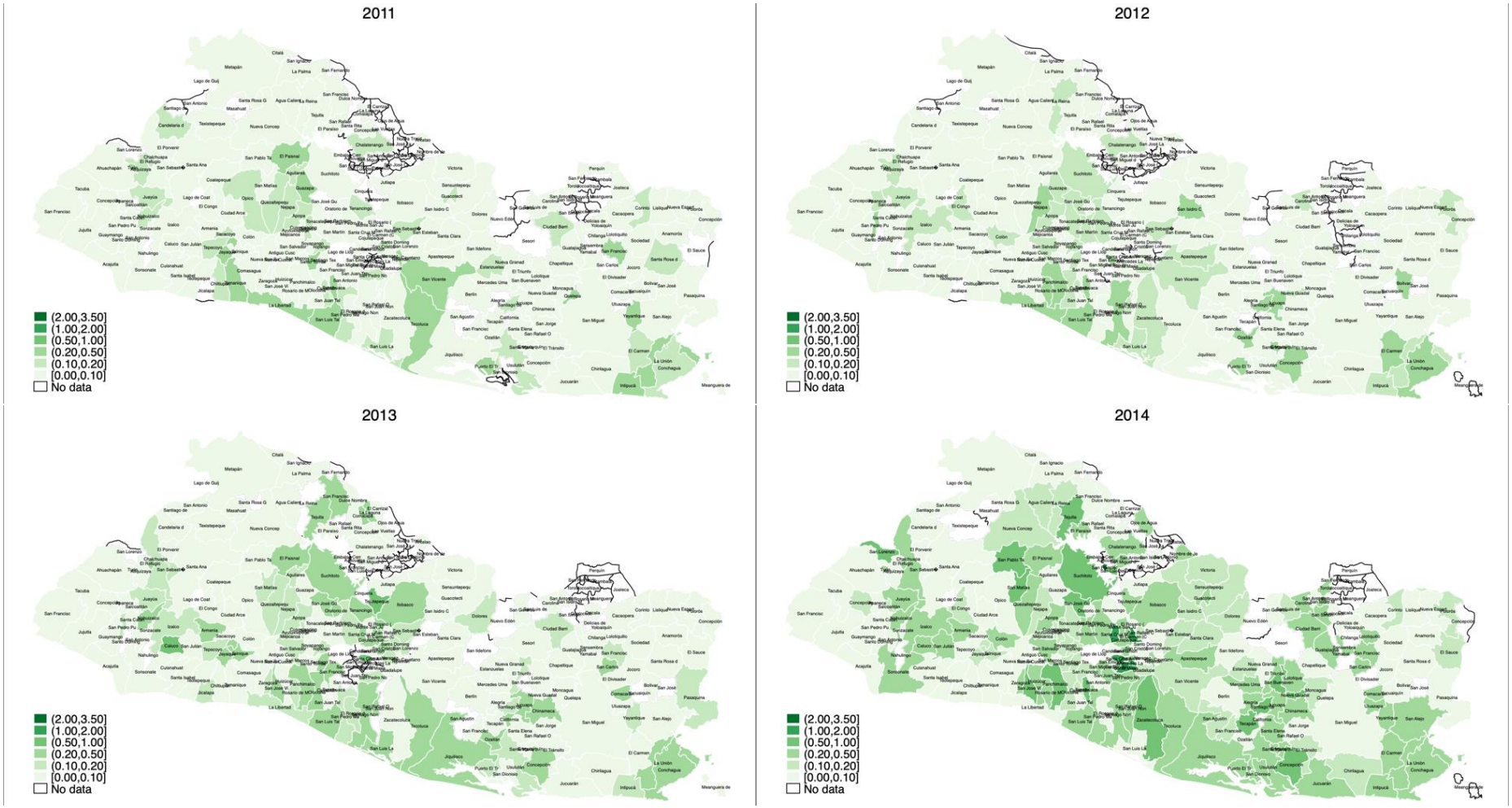
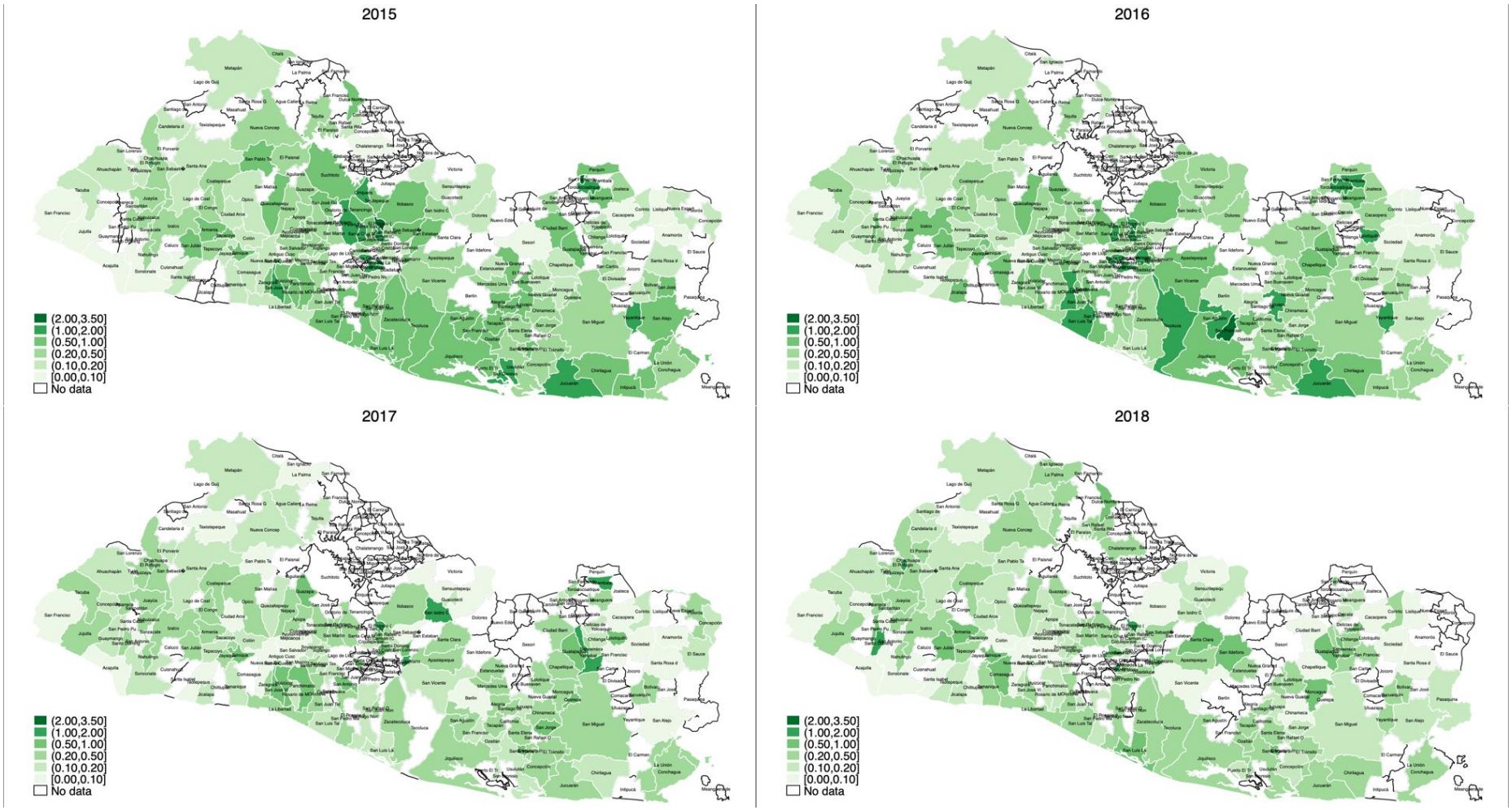
Appendix 1. Homicides Committed Under the Influence of Mara Salvatrucha and Barrio 18 per 1,000 Inhabitants (2011–2020)

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No data
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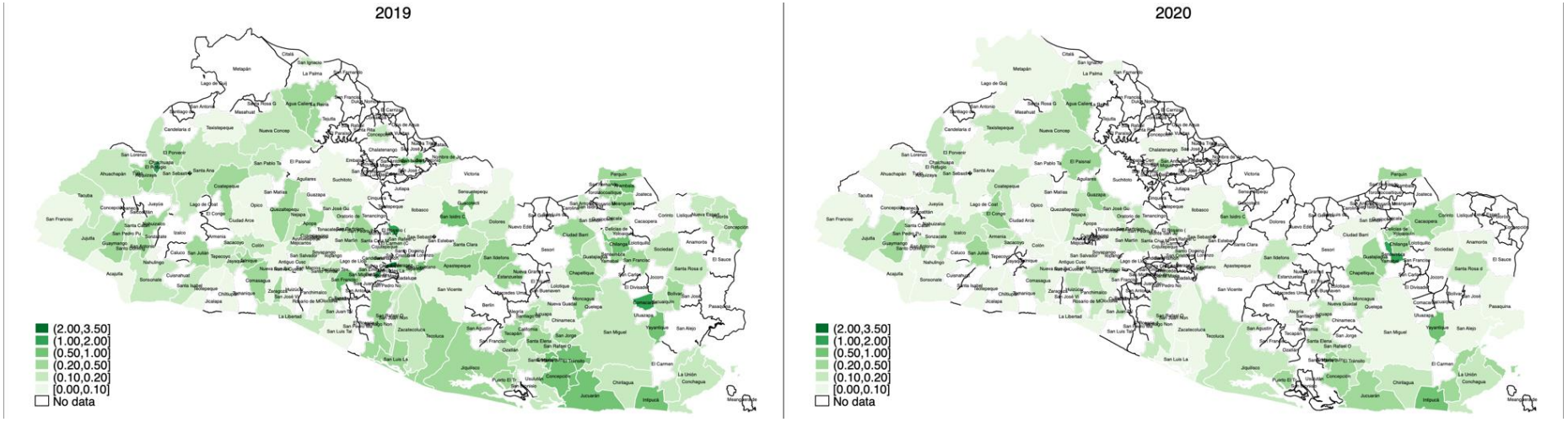

Source: Maps created by the authors.

Appendix 2. Total Homicides Attributed to Gangs per Department, 2010–2020

Department	Homicides by gangs Total / Per 1,000		Homicides by MS-13 Total / Per 1,000	
Ahuachapán	335	0.11	240	0.08
Cabañas	292	0.20	165	0.11
Chalatenango	177	0.09	123	0.07
Cuscatlán	691	0.30	366	0.16
La Libertad	865	0.19	582	0.13
La Paz	768	0.26	357	0.12
La Unión	381	0.17	289	0.13
Morazán	228	0.13	219	0.13
San Miguel	606	0.15	578	0.14
San Salvador	2525	0.16	1034	0.07
San Vincente	331	0.21	252	0.16
Santa Ana	610	0.12	313	0.07
Sonsonate	661	0.16	296	0.07
Usulután	873	0.25	591	0.17
TOTAL	9,343	1.78	5,405	0.11

Source: Authors' estimations based on El Salvador National Police Open Data.
Note: The presence of gangs is measured as the total homicides committed by MS-13 and Barrio 18 per 1,000 inhabitants.

Appendix 3. Definition of Variables and Information Sources

Source	Variable	Description
Fiscal Transparency Portal (Portal de Transparencia Fiscal)	IngCorr	Current revenues (excluding FODES) (US$)
	GastCons	Consumption or operational management expenditures (US$)
	Gasto	Sum of consumption expenditures and financial expenditures
	Tax_rev	Tax revenues (US$)
	NoTax_rev	Non-tax revenues (US$)
	Own_rev	Own revenues, understood as the percentage of tax revenues to total revenues (tax and non-tax)
	Own_rev_pc	Per capita collection, understood as tax revenues per inhabitant
	Cap_Exp	Capital expenditures (US$)
	Current_transfer	Current transfers (US$)
	Capac_build	Investment in human capital (US$), a proxy for the municipality's capabilities
	PC_rev	Per capita income
The Salvadorian Institute of Municipal Development (ISDEM)	FODES	Annual allocation to FODES (US$)
The National Police Open Data	Maras	Homicides committed by Mara Salvatrucha (individuals)
	Other_gangs	Total homicides committed by all gangs
The General Superintendence of Electricity and Telecommunications (SIGET)	Kw	Energy consumption (millions of kWh)
Google Earth Engine	Luminosity	Luminosity
National Institute of Statistics and Census	Popul	Population of the local jurisdiction
The Multi-purpose Household Survey (EHPM)	Density	Inhabitants per km^2
	IncomeH	Family income
	PubService	Percentage of inhabitants who dispose of garbage through a public collector, a proxy for the provision of public services
	Urban_popul	Urban population (individuals)
	Agedpopulation	Percentage of the population over 65 years old

Source: Authors' elaboration.

Appendix 4. Effect of Homicides by All Gangs on Municipalities' Per Capita Own Revenue

Variable	Municipalities			
	All municipalities	Population < 10K	Population 10K–50K	Population > 50K
Other_gangs	-0.114 (0.885)	-0.348 (0.958)	-0.594 (4.118)	-5.327 (10.193)
Kw	0.217*** (0.024)	0.174 (0.261)	0.856*** (0.036)	0.145*** (0.030)
Urban_popul	-0.000* (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
Transf_pc	-0.133*** (0.030)	-0.015 (0.015)	-0.093 (0.077)	0.647* (0.323)
Density	-0.016 (0.132)	-0.068 (0.164)	0.281 (0.960)	6.542 (7.792)
Garbage_sv	2.089 (1.992)	0.476 (1.355)	10.981** (3.867)	16.873 (20.782)
Constant	27.326*** (5.517)	15.239*** (3.247)	8.653 (6.993)	-2.579 (26.948)
Observations	1308	465	646	197

Source: Authors' elaboration

Notes: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix 5. Effect of Homicides by All Gangs on Municipalities' Per Capita Expenditure

Variable	Municipalities			
	All municipalities	Population < 10K	Population 10–50K	Population > 50K
Other_gangs	-18.308*** (2.222)	-18.088*** (2.217)	-4.374 (6.291)	-2.331 (10.745)
Kw	-0.011 (0.020)	-3.000** (1.128)	0.134 (0.217)	-0.034 (0.022)
Urban_popul	-0.000 (0.000)	0.000 (0.001)	-0.000 (0.000)	-0.000** (0.000)
Transf_pc	0.977*** (0.048)	1.046*** (0.054)	0.889*** (0.219)	0.633*** (0.182)
Density	3.212* (1.591)	3.101 (1.675)	-4.119 (2.842)	-22.394 (16.031)
PC_own_rev	0.961*** (0.053)	1.262*** (0.165)	0.791*** (0.144)	1.260*** (0.112)
Constant	32.307*** (4.282)	40.331*** (7.425)	37.311* (15.168)	22.301** (7.450)
Observations	1331	479	655	197

Source: Authors' elaboration

Notes: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix 6. Effect of Homicides by All Gangs on the Municipalities' Quality of Public Services

Variable	Municipalities			
	All municipalities	Population < 10K	Population 10K–50K	Population > 50K
Other_gangs	-0.002* (0.001)	0.004 (0.010)	-0.003* (0.001)	0.000 (0.001)
Urban_popul	0.000*** (0.000)	0.000 (0.000)	0.000** (0.000)	0.000* (0.000)
Households_income	0.000*** (0.000)	0.000** (0.000)	0.000*** (0.000)	-0.000 (0.000)
PC_own_rev	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.001)
Density	0.001 (0.006)	-0.000 (0.007)	0.004 (0.010)	-0.011 (0.029)
MunCapacBuild	0.026 (0.015)	0.049* (0.022)	-0.012 (0.020)	-0.034 (0.019)
Constant	0.188*** (0.042)	0.206** (0.070)	0.082* (0.038)	0.526*** (0.057)
Observations	1391	520	662	209

Source: Authors' elaboration
Notes: Robust standard errors in parentheses. The specification includes year and municipal random effects (not shown). The gang variable is lagged one year, to reduce endogeneity concerns. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix 7. Impact of Gangs on Economic Activity

Variable	Economic activity (proxy change in electricity consumption, kWh)
All_gangs	-3.226*** (0.534)
luminosity	0.012*** (0.001)
PobUrb	0.000** (0.000)
ingfa	0.057** (0.019)
gasto_pc	-0.006 (0.010)
agedpopulation	142691.3 (81298.1)
Constant	-50.060*** (12.051)
Spatial Correlation Outputs WMa	
kw_mm	0.051 (0.074)
All_gangs	-1.272 (0.829)
Observations	144

Source: Authors' elaboration
Note: Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.00$.